Here, $C = 1.6331$ and $ = 1.8749$ are the parameters giving this accuracy.

9. Implement repeated experiments and visualize the average incumbent values and error bars across ten experiments. We first put the code that implements our BayesOpt loop in an outer loop that iterates over multiple experiments. We store the best-seen value at each step across the experiments in `incumbents`.

```
num_repeats = 10

incumbents = torch.zeros((num_repeats, num_queries))

for trial in range(num_repeats):
    print("trial", trial)

    torch.manual_seed(trial) ❶
    train_x = bounds[0] + (bounds[1] - bounds[0]) * torch.rand(1, 2) ❶
    train_y = f(train_x) ❶

    for i in tqdm(range(num_queries)):
        incumbents[trial, i] = train_y.max() ❷

        ... ❸

torch.save(incumbents, [path to file]) ❹
```

❶ Uniformly sample a data point in the search space as the starting point.

❷ Keep track of the best-seen value.

❸ Omitted code is the same as before.

❹ Save results to a file so that we can visualize them later.

We then implement a helper function that plots the average incumbent values and error bars. This function reads in a PyTorch tensor saved at `path`, which should be the saved version of `incumbents` in the previous step

```
def show_agg_progress(path, name):
    def ci(y): ❶
        return 2 * y.std(axis=0) / np.sqrt(num_repeats) ❶

    incumbents = torch.load(path) ❷

    avg_incumbent = incumbents.mean(axis=0) ❸
    ci_incumbent = ci(incumbents) ❸

    plt.plot(avg_incumbent, label=name) ❹
    plt.fill_between( ❹
        np.arange(num_queries), ❹
        avg_incumbent + ci_incumbent, ❹
        avg_incumbent - ci_incumbent, ❹
        alpha=0.3, ❹
    ) ❹
```

❶ Helper sub-function to compute the error bars.

❷ Load saved optimization results.

❸ Compute the mean results and error bars.

❹ Visualize the mean results and error bars. We then can run the policies we have above and compare their performance.

©Manning Publications Co. To comment go to liveBook
https://livebook.manning.com/#!/book/bayesian-optimization-in-action/discussion

Licensed to ashwin sampath <ashwin.manning@gmail.com>